The University of Texas at Austin

Oden Institute for Computational Engineering and Sciences

# Multimodal integration reveals complementary imaging and genetic predictors of coronary artery disease risk

**Devansh Pandey**[1], Liaoyi Xu[1], Eucharist Kun[1], Chenfei Li[1], Joyce Y. Wang[1], Alaa Melek[2], Julie C. DiCarlo[3,5], Edward Castillo[2], Jagat Narula[5], Charles A. Taylor[3,5], Vagheesh M. Narasimhan[1,3,6,†]

[1]Department of Integrative Biology, The University of Texas at Austin · [2]Department of Biomedical Engineering, The University of Texas at Austin · [3]Oden Institute for Computational Engineering and Sciences, The University of Texas at Austin · [4]University of Texas Health Sciences Center, Houston · [5]Department of Internal Medicine, Dell Medical School, The University of Texas at Austin · [6]Department of Statistics and Data Science, The University of Texas at Austin · [†]Corresponding author · Presenting author in bold

**Cardiac imaging and polygenic risk capture complementary information about future CAD, improving risk stratification beyond clinical variables in UK Biobank.**

## 1 Cardiac MRI carries the imaging signal

**AUC 0.77 vs 0.67**

learned embeddings vs conventional phenotypes; long-axis and aortic MRI add most

## 2 Imaging and genetic risk barely overlap

**r = 0.065**

imaging risk score vs polygenic score; shared variance 0.4%

## 3 Combining them sharpens who is at risk

**C-index 0.819 vs 0.749**

full vs clinical model; 5-year incidence 15.9% in the top tier

## How the study was built

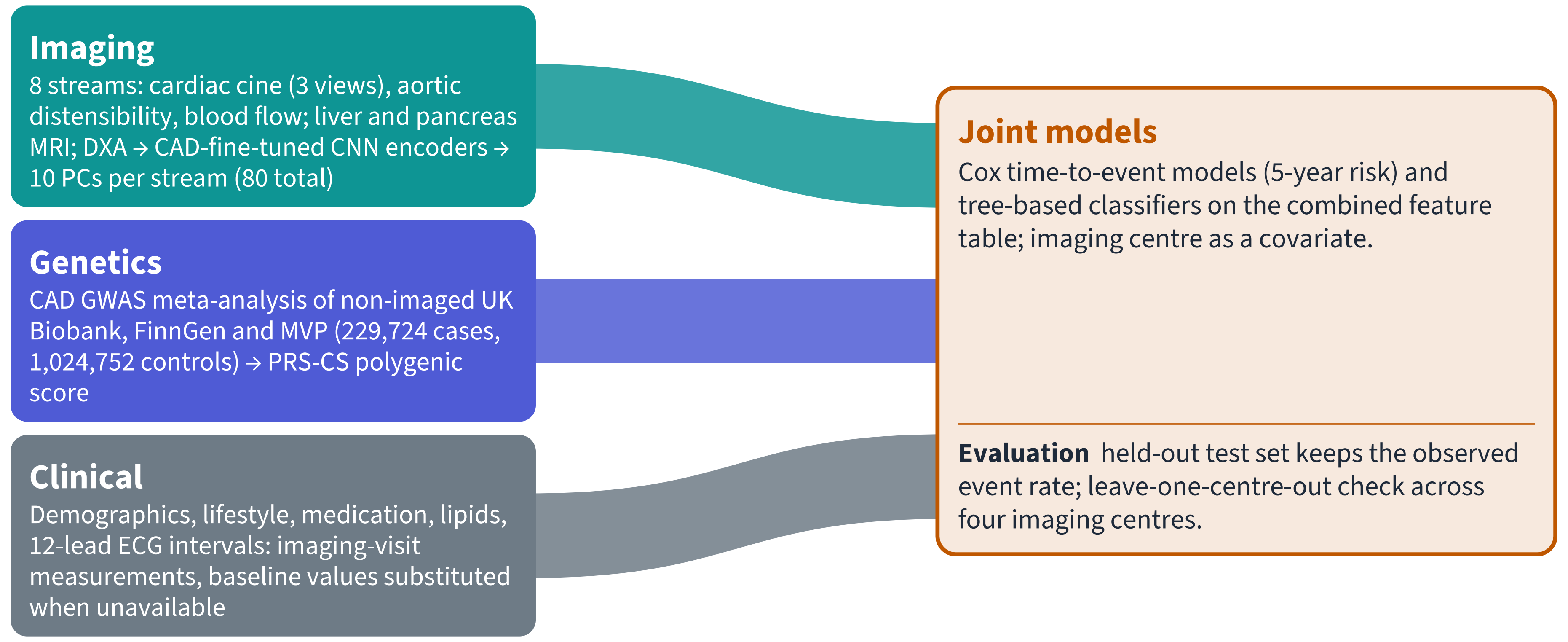
**42,351**
UK Biobank participants with cardiac and abdominal MRI, DXA, ECG, genotypes and labs

**677**
incident CAD cases after the imaging visit (1.6% observed event proportion)

**4.6 years**
median follow-up; all risk models target 5-year risk

**75 / 25**
spatially stratified split: 31,754 train (499 events) / 10,597 held-out (178 events)

### Three data streams, one model



**Imaging**
8 streams: cardiac cine (3 views), aortic distensibility, blood flow; liver and pancreas MRI; DXA → CAD-fine-tuned CNN encoders → 10 PCs per stream (80 total)

**Genetics**
CAD GWAS meta-analysis of non-imaged UK Biobank, FinnGen and MVP (229,724 cases, 1,024,752 controls) → PRS-CS polygenic score

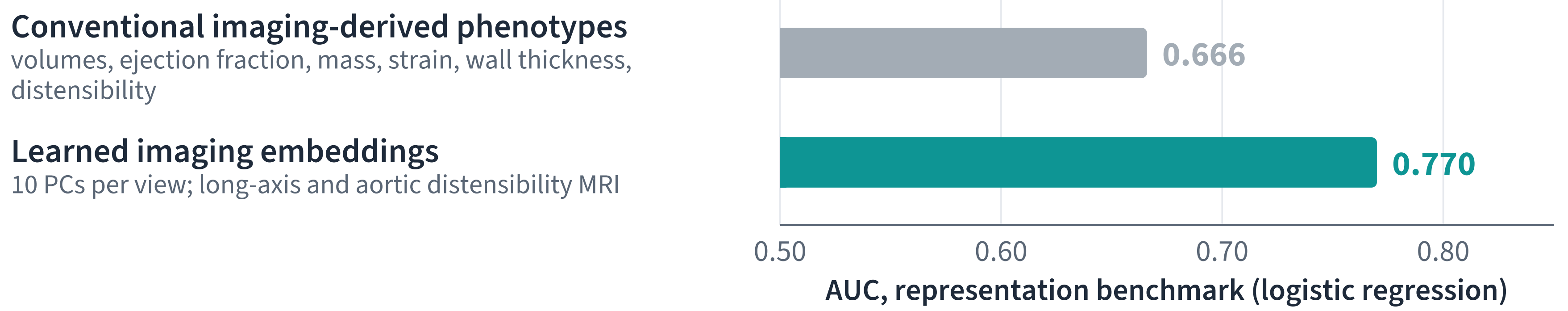
**Clinical**
Demographics, lifestyle, medication, lipids, 12-lead ECG intervals: imaging-visit measurements, baseline values substituted when unavailable

**Joint models**
Cox time-to-event models (5-year risk) and tree-based classifiers on the combined feature table; imaging centre as a covariate.

**Evaluation** held-out test set keeps the observed event rate; leave-one-centre-out check across four imaging centres.

**Outcome** clinically recognised CAD (myocardial infarction, angina or ischaemic heart disease, coronary revascularisation) from linked records; prevalent CAD excluded. Cardiac MRI does not image coronary plaque. Encoders (Med3D 3D ResNet-18 for cardiac MRI; RadImageNet 2D ResNet-50 for liver, pancreas and DXA) were fine-tuned on balanced case–control sets from the training partition only; all risk analyses keep the observed event rate. Genetically defined White British participants.

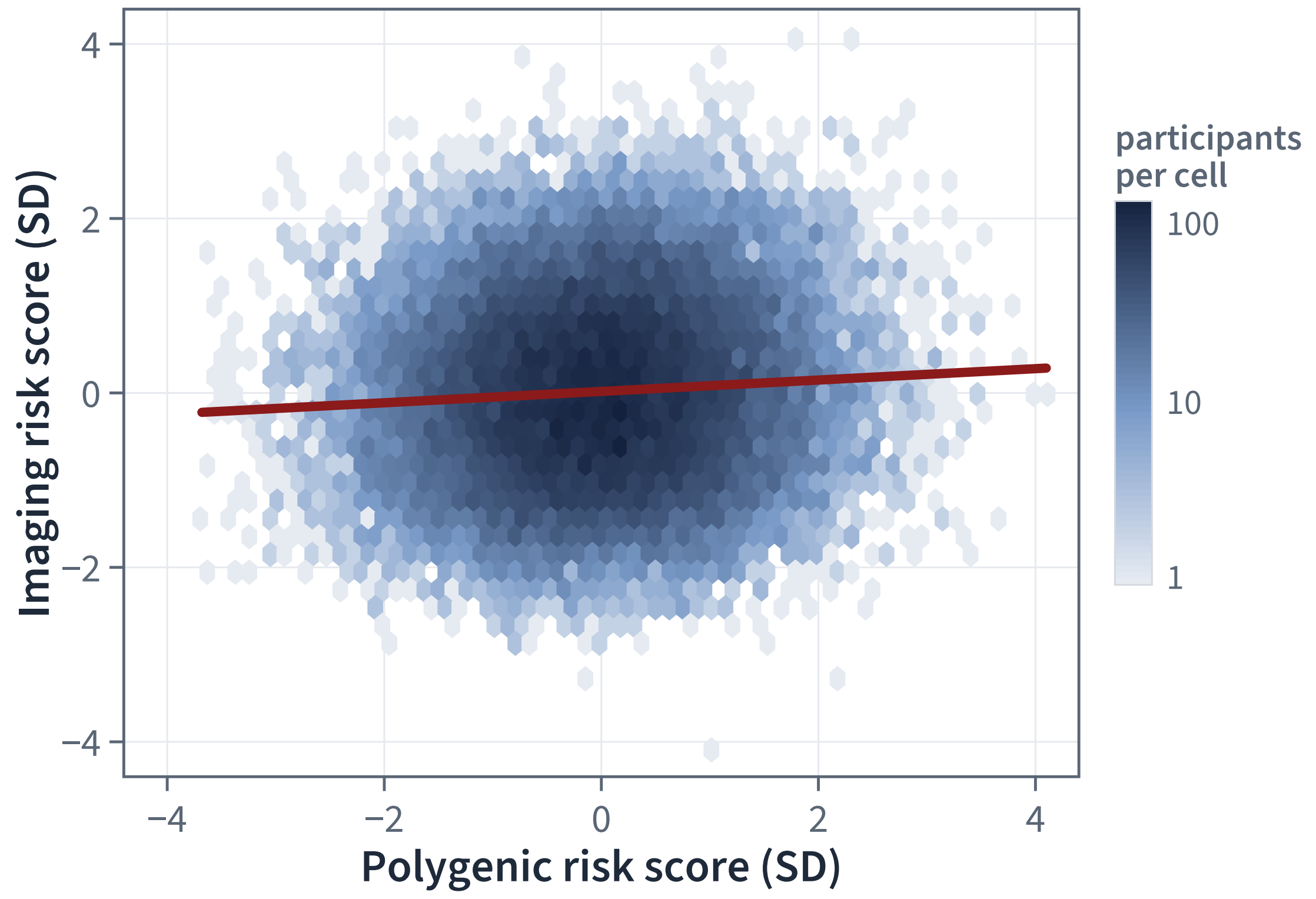
### Learned representations beat conventional phenotypes



Separate from the held-out survival analyses. Leading canonical components of the embeddings align with stroke volume, ejection fraction and myocardial mass (canonical correlations 0.79–0.84).

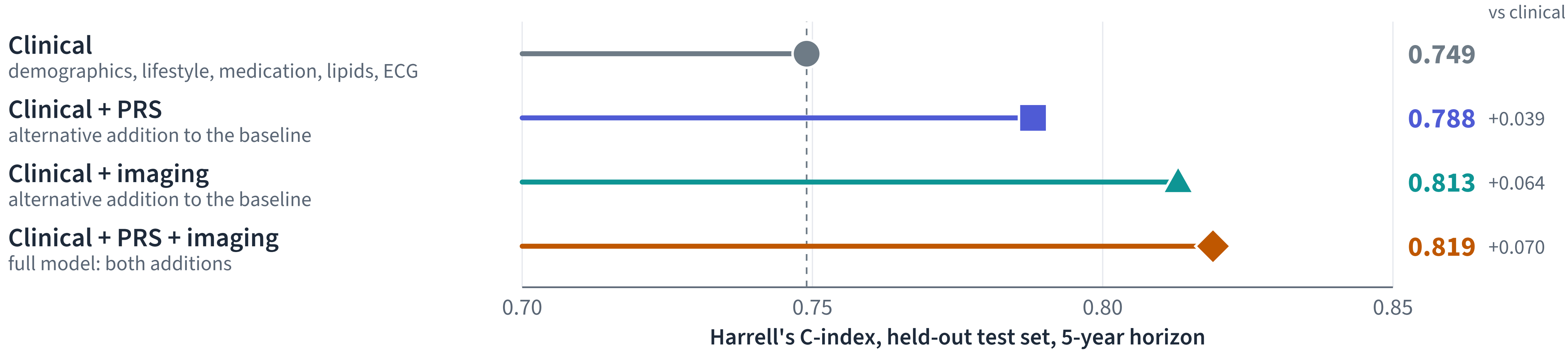
## Imaging and genetics see different risk

**r = 0.065.** The imaging risk score (ridge Cox model on the 80 imaging PCs, five-fold cross-fitted) and the polygenic score share 0.4% of their variance (n = 29,075, training partition). Both predict CAD; they rank different people.



Low overlap supports complementarity; it does not establish biological independence or show that imaging risk is wholly acquired. Genetics explains 0.15% of the imaging risk score; demographics explain 38.4%.

### Held-out discrimination: the full model reaches C-index 0.819



Nested Cox models; n = 10,597, 178 events. Differences are absolute. No confidence intervals reported. Calibration slopes 0.99–1.08.

**Largest loss of explained variance when removed**

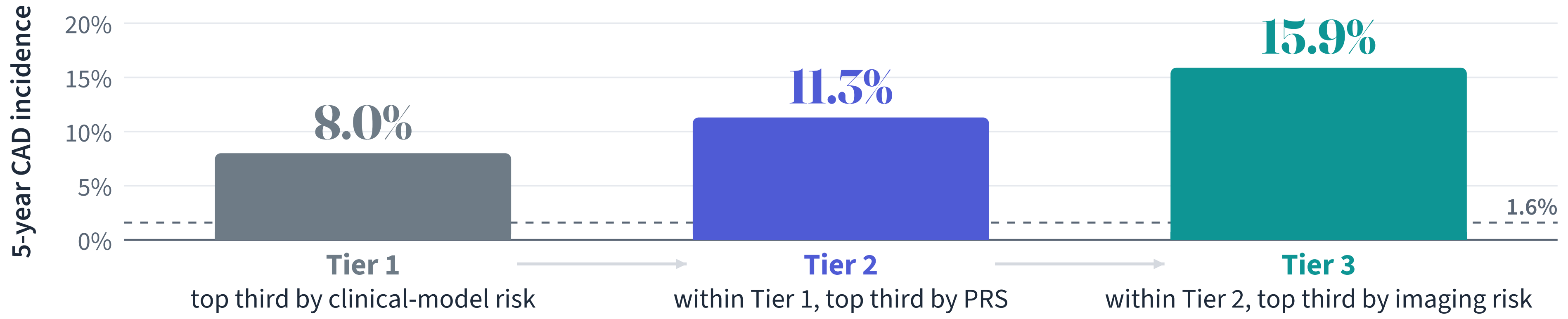
- Long-axis cardiac MRI
- Medication history
- Polygenic risk score
- Aortic distensibility MRI

**Little loss after clinical adjustment**

- Liver MRI
- Pancreas MRI
- Whole-body DXA
- Blood-flow MRI

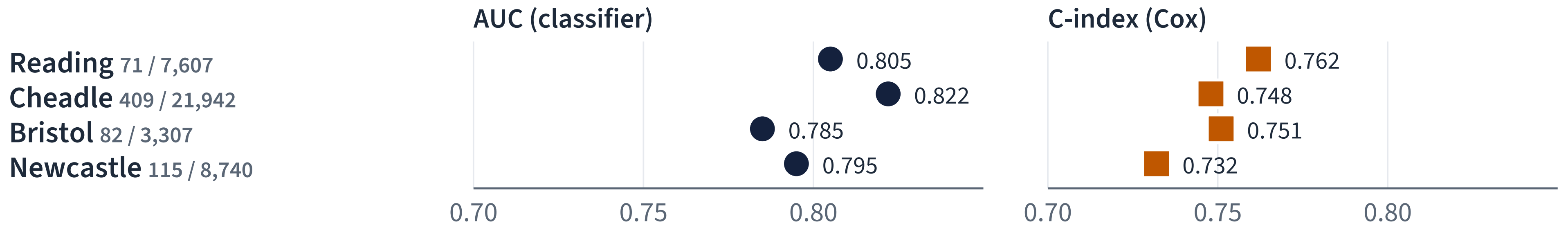
## Combined models find the highest-risk people

Sequential stratification of the held-out set: each tier keeps the top third of the previous tier, with the cut-point recalculated within that tier.



Dashed line: 1.6% observed event proportion in the full cohort (crude, not a 5-year estimate). Within Tier 3, the top within-tier quintile of both PRS and imaging risk reached 36.8% five-year incidence; stratum size and confidence interval are not reported.

### Stable across imaging centres (leave-one-centre-out, within UK Biobank)



Rows: held-out centre (test cases / controls). Site-analysis models trained on three centres, tested on the fourth. Different feature set and validation design from the main held-out analysis; not external validation.

**0.634**
continuous NRI, full vs clinical (95% CI 0.534–0.734)

**5.5 to 9.5%**
index of prediction accuracy, clinical to full model

**0.99–1.08**
calibration slopes, all nested models

### Conclusion

Selected cardiac imaging (long-axis cine and aortic distensibility MRI) and genome-wide polygenic risk add partially non-overlapping information to clinical variables, improving discrimination and risk enrichment, with calibration slopes near 1.

### Limitations

- Selected, complete-case, White British UK Biobank sample; absolute risks may not transport.
- No fully external validation; site robustness is within one cohort.
- No coronary calcium or CT-angiography benchmark; cardiac MRI does not image plaque.
- Observational prediction: clinical utility, test ordering and cost-effectiveness untested.
- Extreme-risk strata are small and their precision is not reported.

---

**Data and code**

UK Biobank data are available to approved researchers through application (this work: application 65439). Analysis and figure code: github.com/Devanshpandey/Multimodal-integration-for-coronary-artery-disease-risk-prediction (code only; no participant data or model weights). Genome-wide polygenic risk score; no individual gene-level discovery claims.

**Preprint (earlier version)**

medRxiv doi:10.1101/2025.10.09.25337677

**Acknowledgements**

Computing: Texas Advanced Computing Center (Director's Discretionary Fund). D.P.: Outstanding Graduate Research Fellowship, The Graduate School, UT Austin. V.M.N.: Allen Discovery Center program (Paul G. Allen Frontiers Group, Paul G. Allen Family Foundation) and Good Systems Fellowship for Ethical AI, UT Austin.

**Disclosures**

C.A.T. founded HeartFlow, Inc. and serves on the boards of HeartFlow, Inc. and Ebenbuild GmbH. The other authors declare no competing interests.